Homework Solution
Chapter 5: Present Worth and Related Economic Criteria

Instructor: Naim Al Khoury
Teaching Assistant: Mohammad Younis

October 2, 2026

Exercise 1: Classifying Alternatives

Solution:

Part 1. The two transformer systems are **mutually exclusive**, since only one can be installed (there is only space and budget for a single unit) and choosing one automatically rejects the other. The three grid upgrades are **independent projects**, since each can be approved or rejected on its own merits, and any combination of them (including all three, or none) can be implemented together.

Part 2. Since both groups only generate cost savings and no new revenue, both the transformer alternatives and the three independent upgrades should be treated as **cost (service) alternatives**. For cost alternatives, the objective is to minimize the PW of costs rather than maximize a profit-based measure.

Exercise 2: The AI Data Center Dilemma

Solution:

Step 1: Interest rate and time unit. Since the cash flows are monthly, the analysis is done on a monthly basis:

$$i = \frac{12\%}{12} = 1\% \text{ per month}, \qquad N = 60 \text{ months}$$

Step 2: Cash flows.

- Initial investment: \$4,800,000 at $t = 0$.

- Fluid swap: \$150,000 every 2 years, i.e., at $t = 24$ and $t = 48$ monthse.
- Maintenance: \$15,000 per month for months 1–12; from month 13 onward, \$16,000 at month 13 increasing by $G = \$1{,}000$ per month (so month 60 costs \$63,000).
- Energy savings: \$80,000 per month for months 1–60.

Step 3: Present worth of each component.

Factors:

$$(P/A, 1\%, 12) = 11.2551, \quad (P/A, 1\%, 48) = 37.9740, \quad (P/A, 1\%, 60) = 44.9550$$
$$(P/G, 1\%, 48) = 820.1460, \quad (P/F, 1\%, 12) = 0.8874$$
$$(P/F, 1\%, 24) = 0.7876, \quad (P/F, 1\%, 48) = 0.6203$$

Fluid swaps:

$$\begin{aligned} PW_{\text{fluid}} &= -150{,}000\,[(P/F, 1\%, 24) + (P/F, 1\%, 48)] \\ &\approx -\$211{,}173.98 \end{aligned}$$

Maintenance, months 1–12 (uniform series):

$$\begin{aligned} PW_{M1} &= -15{,}000(P/A, 1\%, 12) \\ &\approx -\$168{,}826.16 \end{aligned}$$

Maintenance, months 13–60: this is a \$16,000 base amount (beginning at month 13) plus a gradient of \$1,000 whose first increase appears at month 14 ($n = 48$ periods measured from month 12), both discounted back 12 months:

$$\begin{aligned} PW_{M2} &= [-16{,}000(P/A, 1\%, 48) - 1{,}000(P/G, 1\%, 48)]\,(P/F, 1\%, 12) \\ &\approx -\$539{,}199.37 - \$727{,}837.94 \\ &= -\$1{,}267{,}037.31 \end{aligned}$$

Energy savings:

$$\begin{aligned} PW_{\text{sav}} &= 80{,}000(P/A, 1\%, 60) \\ &\approx \$3{,}596{,}403.07 \end{aligned}$$

Step 4: Total present worth.

$$\begin{aligned} PW = &-4{,}800{,}000 - 211{,}173.98 - 168{,}826.16 \\ &- 1{,}267{,}037.31 + 3{,}596{,}403.07 \\ \approx &-\$2{,}850{,}634.38 \end{aligned}$$

Decision: Since $PW \approx -\$2{,}850{,}634.38 < 0$, the project does not earn the 12% (nominal, compounded monthly) MARR. **The company should not invest** in the liquid immersion cooling system.

Exercise 3: Independent Projects Screening with Capital Rationing

Solution:

Part 1. Unlimited budget: each project is evaluated against the Do-Nothing alternative at $i = 9\%$; every project with $PW \geq 0$ is accepted.

Project 1 (perpetuity, capitalized cost formula $P = A/i$):

$$\begin{aligned} PW_1 &= -200{,}000 + \frac{25{,}000}{0.09} \\ &\approx \$77{,}777.78 \end{aligned}$$

Project 2, using $(P/A, 9\%, 10) = 6.4177$:

$$\begin{aligned} PW_2 &= -300{,}000 + 65{,}000(P/A, 9\%, 10) \\ &\approx \$117{,}147.75 \end{aligned}$$

Project 3 (geometric gradient with $A_1 = 15{,}000$, $g = 6\%$, $i = 9\%$, $n = 10$). Since $g \neq i$:

$$\begin{aligned}(P/A, g, i, n) &= \frac{1 - \left(\frac{1+g}{1+i}\right)^n}{i - g} \\ &= \frac{1 - \left(\frac{1.06}{1.09}\right)^{10}}{0.09 - 0.06} \\ &= \frac{1 - 0.7565}{0.03} \approx 8.1177 \\ PW_3 &= -150{,}000 + 15{,}000(8.1177) \\ &\approx -\$28{,}236.71\end{aligned}$$

Since only Projects 1 and 2 have $PW > 0$, **the cooperative should implement Projects 1 and 2** and reject Project 3 (total initial cost \$500,000).

Part 2. With a \$400,000 budget, Projects 1 and 2 together (\$500,000) are no longer feasible. Project 3 has a negative PW and should never be selected on its own merits, but it must still be checked within feasible bundles. The feasible bundles (total cost $\leq$ \$400,000) are:

Bundle	**Initial cost**	**Total PW**
Do-nothing	\$0	\$0.00
Project 1 only	\$200,000	\$77,777.78
Project 2 only	\$300,000	\$117,147.75
Project 3 only	\$150,000	−\$28,236.71
Projects 1 + 3	\$350,000	\$49,541.07

(Bundles 1 + 2, 2 + 3, and 1 + 2 + 3 exceed the budget.)

The optimal choice is to implement Project 2 only, since it has the highest total PW (\$117,147.75) among all feasible bundles.

Exercise 4: Mutually Exclusive Revenue Alternatives with Equal Lives

Solution:

Using $(P/A, 10\%, 5) = 3.7908$ and $(P/F, 10\%, 3) = 0.7513$:

$$\begin{aligned} PW_X &= -250{,}000 + 58{,}000(P/A, 10\%, 5) \\ &\approx -\$30{,}134.37 \end{aligned}$$

$$\begin{aligned} PW_Y &= -140{,}000 + 34{,}000(P/A, 10\%, 5) \\ &\quad - 40{,}000(P/F, 10\%, 3) \\ &\approx -\$41{,}165.84 \end{aligned}$$

Both alternatives have a negative PW at the 10% MARR, so neither meets the minimum required return. Per the PW decision rule for mutually exclusive revenue alternatives (if no alternative has $PW \geq 0$, Do-Nothing is selected), **the company should select the Do-Nothing alternative** and continue with the current manual monitoring process.

Exercise 5: Comparing Evaluation Methods: LCM vs. Study Period

Solution:

Part 1: LCM approach. The LCM of 2 and 3 is **6 years**: System A is purchased 3 times ($t = 0, 2, 4$) and System B twice ($t = 0, 3$). Factors at 10%:

$$(P/A, 10\%, 6) = 4.3553, \qquad (P/F, 10\%, 2) = 0.8264, \qquad (P/F, 10\%, 3) = 0.7513$$
$$(P/F, 10\%, 4) = 0.6830, \qquad (P/F, 10\%, 6) = 0.5645$$

System A cash flows: $-150{,}000$ at $t = 0, 2, 4$ (initial costs, the reinvestments at $t = 2, 4$ coincide with salvage at those times); salvage $+15{,}000$ at $t = 2, 4, 6$; savings $+115{,}000$ at $t = 1, \ldots, 6$.

$$\begin{aligned} PW_A &= -150{,}000 - 150{,}000(P/F, 10\%, 2) \\ &\quad - 150{,}000(P/F, 10\%, 4) + 115{,}000(P/A, 10\%, 6) \\ &\quad + 15{,}000\left[(P/F, 10\%, 2) + (P/F, 10\%, 4) + (P/F, 10\%, 6)\right] \\ &\approx \$155{,}545.02 \end{aligned}$$

System B cash flows: $-250{,}000$ at $t = 0, 3$; salvage $+30{,}000$ at $t = 3, 6$; savings $+120{,}000$ at $t = 1, \ldots, 6$.

$$\begin{aligned} PW_B &= -250{,}000 - 250{,}000(P/F, 10\%, 3) \\ &\quad + 120{,}000(P/A, 10\%, 6) \\ &\quad + 30{,}000\left[(P/F, 10\%, 3) + (P/F, 10\%, 6)\right] \\ &\approx \$124{,}276.25 \end{aligned}$$

Since $PW_A > PW_B > 0$, **System A should be selected** under the LCM approach.

Part 2: 3-year study period. System B's life equals the study period, so its PW over 3 years is evaluated directly, using $(P/A, 10\%, 3) = 2.4869$ and $(P/F, 10\%, 3) = 0.7513$:

$$\begin{aligned} PW_B &= -250{,}000 + 120{,}000(P/A, 10\%, 3) \\ &\quad + 30{,}000(P/F, 10\%, 3) \\ &\approx \$70{,}961.68 \end{aligned}$$

System A's life (2 years) is shorter than the study period, so its net cash flows are reinvested at the MARR (10%) in Year 3. The net cash flow at $t = 2$ is $115{,}000 + 15{,}000 = 130{,}000$.

$$\begin{aligned} FW_{A,3} &= 115{,}000(F/P, 10\%, 2) + 130{,}000(F/P, 10\%, 1) \\ &= 115{,}000(1.21) + 130{,}000(1.10) \\ &= \$282{,}150 \\ PW_A &= -150{,}000 + 282{,}150(P/F, 10\%, 3) \\ &\approx \$61{,}983.47 \end{aligned}$$

Since $PW_B > PW_A > 0$, **System B should be selected** under the 3-year study period approach.

Part 3. Yes, the decision changes (A under LCM, B under the study period). The LCM approach implicitly assumes that each system can be repeatedly replaced with identical cash flows, so System A's shorter life is fully exploited through repeated high-return reinvestments in new units over 6 years. The study period approach instead truncates the evaluation at 3 years and assumes the money earned by System A is only reinvested at the MARR (a return of zero NPV) for the remaining year, so A's advantage of repeated replacement is lost, while B's higher savings and salvage count fully. Different assumptions about what happens after the shorter life (replication vs. reinvestment at

the MARR) therefore lead to different rankings.

Exercise 6: Comparing Alternatives with Unequal Lives (LCM Approach)

Solution:

Part 1: Equal-service period. The LCM of 4 and 6 is **12 years**: Truck P is purchased 3 times ($t = 0, 4, 8$) and Truck Q twice ($t = 0, 6$).

Part 2: Cash flows over 12 years.

Truck P: cost \$180,000 at $t = 0, 4, 8$; savings \$52,000 for $t = 1, \ldots, 12$; no salvage.

Truck Q: cost \$260,000 at $t = 0, 6$; savings \$68,000 for $t = 1, \ldots, 12$; salvage \$20,000 at $t = 6, 12$; engine overhaul \$30,000 at the end of Year 3 of each life cycle, i.e., at $t = 3$ and $t = 9$.

Part 3: Present worth at $i = 9\%$. Factors:

$$\begin{aligned}
(P/F, 9\%, 3) &= 0.7722, \quad (P/F, 9\%, 4) = 0.7084, \quad (P/F, 9\%, 6) = 0.5963 \\
(P/F, 9\%, 8) &= 0.5019, \quad (P/F, 9\%, 9) = 0.4604, \quad (P/F, 9\%, 12) = 0.3555 \\
(P/A, 9\%, 12) &= 7.1607
\end{aligned}$$

$$\begin{aligned}
PW_P &= -180{,}000 - 180{,}000(P/F, 9\%, 4) \\
&\quad - 180{,}000(P/F, 9\%, 8) + 52{,}000(P/A, 9\%, 12) \\
&\approx -\$25{,}494.75
\end{aligned}$$

$$\begin{aligned}
PW_Q &= -260{,}000 - 260{,}000(P/F, 9\%, 6) \\
&\quad + 20{,}000\,[(P/F, 9\%, 6) + (P/F, 9\%, 12)] \\
&\quad - 30{,}000\,[(P/F, 9\%, 3) + (P/F, 9\%, 9)] \\
&\quad + 68{,}000(P/A, 9\%, 12) \\
&\approx \$53{,}957.52
\end{aligned}$$

Since $PW_Q > 0 > PW_P$, **the company should purchase Truck Model Q**.

Exercise 7: Evaluating Alternatives Over a Fixed Study Period

Solution:

The study period is 6 years at $i = 10\%$. Factors: $(P/A, 10\%, 6) = 4.3553$, $(P/F, 10\%, 6) = 0.5645$, $(P/A, 10\%, 4) = 3.1699$, $(P/F, 10\%, 4) = 0.6830$.

Machine Alpha (life 8 years > study period 6 years): the salvage value is replaced by the estimated market value at the end of Year 6 ($900,000), since the machine still has unused value at the end of the study period.

$$\begin{aligned} PW_\alpha &= -3{,}200{,}000 + 650{,}000(P/A, 10\%, 6) \\ &\quad + 900{,}000(P/F, 10\%, 6) \\ &\approx \$138{,}945.99 \end{aligned}$$

Machine Beta (life 4 years < study period 6 years, not replaced): its net cash flows over Years 1–4 are reinvested at the MARR for the remaining 2 years, which leaves the PW unchanged:

$$\begin{aligned} FW_{\beta,4} &= 400{,}000(F/A, 10\%, 4) + 150{,}000 \\ &= 400{,}000(4.6410) + 150{,}000 \\ &= \$2{,}006{,}400 \\ FW_{\beta,6} &= 2{,}006{,}400(F/P, 10\%, 2) \\ &= 2{,}006{,}400(1.21) \\ &= \$2{,}427{,}744 \\ PW_\beta &= -1{,}400{,}000 + 2{,}427{,}744(P/F, 10\%, 6) \\ &\approx -\$29{,}601.80 \end{aligned}$$

(Equivalently, $PW_\beta = -1{,}400{,}000 + 400{,}000(P/A, 10\%, 4) + 150{,}000(P/F, 10\%, 4)$.)

Decision: Since $PW_\alpha > 0 > PW_\beta$, **the plant manager should select Machine Alpha**; Machine Beta fails to meet the 10% MARR and is rejected.

Exercise 8: Future Worth Analysis: Venture Capital Exit Strategy

Solution:

Step 1: Net cash flows ($i = 18\%$, exit at $N = 6$). Years 3–6 profits grow geometrically at 15% from the Year-2 profit of $1,500,000:

Year	Components	Net cash flow
0	Acquisition	−\$10,000,000
1	$-2{,}000{,}000 - 1{,}000{,}000$	−\$3,000,000
2	$-2{,}000{,}000 + 1{,}500{,}000$	−\$500,000
3	$1{,}500{,}000(1.15)$	\$1,725,000
4	$1{,}500{,}000(1.15)^2$	\$1,983,750
5	$1{,}500{,}000(1.15)^3$	\$2,281,312.50
6	$1{,}500{,}000(1.15)^4$	\$2,623,509.38

Step 2: Future worth of the cash flows at $t = 6$ before the sale, using $(F/P, 18\%, n) = 1.18^n$:

$$\begin{aligned} FW_{0\text{–}2} &= -10{,}000{,}000(F/P, 18\%, 6) \\ &\quad - 3{,}000{,}000(F/P, 18\%, 5) \\ &\quad - 500{,}000(F/P, 18\%, 4) \\ &= -10{,}000{,}000(2.6996) \\ &\quad - 3{,}000{,}000(2.2878) - 500{,}000(1.9388) \\ &\approx -\$34{,}828{,}203.68 \end{aligned}$$

The geometric series (Years 3–6, $A_1 = 1{,}725{,}000$, $g = 15\%$, $i = 18\%$, $n = 4$) has present worth at $t = 2$ of

$$\begin{aligned} P_2 &= A_1 \frac{1 - \left(\frac{1.15}{1.18}\right)^4}{0.18 - 0.15} \\ &= 1{,}725{,}000 \frac{1 - 0.9021}{0.03} \\ &\approx \$5{,}628{,}216.94 \end{aligned}$$

and is carried forward 4 years:

$$\begin{aligned} FW_{3\text{–}6} &= 5{,}628{,}216.94(F/P, 18\%, 4) \\ &= 5{,}628{,}216.94(1.9388) \\ &\approx \$10{,}911{,}861.83 \end{aligned}$$

(which equals the sum of the individual compounded flows: $2{,}834{,}230.20 + 2{,}762{,}173.50 + 2{,}691{,}948.75 + 2{,}623{,}509.38$).

Step 3: Break-even terminal value. Setting $FW = 0$ (exactly achieving the MARR):

$$\begin{aligned} -34{,}828{,}203.68 + 10{,}911{,}861.83 + V &= 0 \\ V &= \$23{,}916{,}341.86 \end{aligned}$$

Thus the firm must sell the startup for **at least \$23,916,341.86** at the end of Year 6.

Exercise 9: Capitalized Cost: Comparing Finite vs. Infinite Facilities

Solution:

Option A: Permanent Suspension Bridge ($i = 6\%$). All capitalized costs are computed as costs ($-$).

Non-recurring cost:

$$CC_{\text{initial}} = -\$4{,}500{,}000$$

Maintenance: a perpetual \$45,000/year from Year 1, plus an additional \$25,000/year perpetuity starting in Year 11 (valued at $t = 10$ and discounted with $(P/F, 6\%, 10) = 0.5584$):

$$\begin{aligned} CC_{\text{base}} &= \frac{-45{,}000}{0.06} = -\$750{,}000.00 \\ CC_{\text{extra}} &= \frac{-25{,}000}{0.06}(P/F, 6\%, 10) \\ &\approx -\$232{,}664.49 \\ CC_{\text{maintenance}} &= -750{,}000.00 - 232{,}664.49 \\ &= -\$982{,}664.49 \end{aligned}$$

Reinforcement every 20 years ($(A/F, 6\%, 20) = 0.02718$):

$$\begin{aligned} A_{\text{reinf}} &= 600{,}000(A/F, 6\%, 20) \\ &\approx \$16{,}310.73 \\ CC_{\text{reinf}} &= \frac{-16{,}310.73}{0.06} \\ &\approx -\$271{,}845.57 \end{aligned}$$

Lighting/safety rails every 25 years ($(A/F, 6\%, 25) = 0.01823$):

$$\begin{aligned} A_{\text{light}} &= 250{,}000(A/F, 6\%, 25) \\ &\approx \$4{,}556.68 \\ CC_{\text{light}} &= \frac{-4{,}556.68}{0.06} \\ &\approx -\$75{,}944.66 \end{aligned}$$

Total:

$$\begin{aligned} CC_A = &-4{,}500{,}000 - 982{,}664.49 \\ &- 271{,}845.57 - 75{,}944.66 \\ \approx &-\$5{,}830{,}454.72 \end{aligned}$$

Option B: Modular Steel Bridge (15-year life, assumed to be replicated indefinitely). Compute the equivalent annual worth over one life cycle, using $(A/P, 6\%, 15) = 0.10296$ and $(A/F, 6\%, 15) = 0.04296$:

$$\begin{aligned} AW_B &= -1{,}800{,}000(A/P, 6\%, 15) - 120{,}000 \\ &\quad + 150{,}000(A/F, 6\%, 15) \\ &\approx -185{,}332.98 - 120{,}000 + 6{,}444.41 \\ &= -\$298{,}888.56 \\ CC_B &= \frac{AW_B}{i} = \frac{-298{,}888.56}{0.06} \\ &\approx -\$4{,}981{,}476.01 \end{aligned}$$

Decision: Since $|CC_B| = \$4{,}981{,}476.01 < |CC_A| = \$5{,}830{,}454.72$, **Option B (the Modular Steel Bridge) is the most economical choice** over an infinite planning horizon.

Exercise 10: Payback Period and Its Limitations

Solution:

Part 1. Using $n_P = P/NCF$:

$$n_{P,M} = \frac{60{,}000}{15{,}000} = 4.00 \text{ years}$$
$$n_{P,N} = \frac{60{,}000}{11{,}000} \approx 5.45 \text{ years}$$

Based strictly on payback period, **Oven M** appears preferred, since it recovers the initial investment faster (4.00 vs. approximately 5.45 years).

Part 2. No, this is not necessarily the most economical choice. Total undiscounted cash received over the full 6-year life:

$$\begin{aligned} \text{Oven M:} \quad & 15{,}000 \times 6 = \$90{,}000 \\ \text{Oven N:} \quad & (11{,}000 \times 6) + 28{,}000 \\ & = 66{,}000 + 28{,}000 = \$94{,}000 \end{aligned}$$

Oven N generates \$4,000 more total cash than Oven M despite its longer payback period. This illustrates the central weakness of payback period analysis: it **completely disregards cash flows occurring after the payback period is reached**, in particular Oven N's \$28,000 salvage value at Year 6. Payback should therefore only be used as a quick, supplemental screening tool, and never as the sole basis for an accept/reject decision.

Multiple Choice Questions

1. **(b)** A/i. Once a finite-life alternative is converted to its equivalent uniform annual worth A (assuming replication forever), the capitalized cost is the perpetuity value A/i.

2. **(c)** The cash flow estimates are the exact same in every subsequent life cycle. The LCM approach assumes each alternative is replicated with identical cash flows (and costs).

3. **(b)** At the end of period 1. Excel's NPV function discounts `value1` by one full period, so an initial cost at $t = 0$ must be added outside the function.

4. **(c)** Each project is evaluated independently against the do-nothing alternative. Independent projects do not compete with each other and can be either revenue or

cost projects.

5. **(d)** Reinvest the accumulated net cash flows at the MARR until the end of the study period.